HEALTH TECHNOLOGY ECOSYSTEM

Request for Information

SUBMITTED TO:

DEPARTMENT OF HEALTH & HUMAN SERVICES (HHS)
Centers for Medicare & Medicaid Services (CMS);

Assistant Secretary for Technology Policy (ASTP)/Office of the National Coordinator for Health Information Technology (ONC) (collectively, ASTP/ONC)

SUBMITTED BY:

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Electronic

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1.0 Organization Information

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Executive Summary - Peraton's Commitment to the Federal Government and the Healthcare Ecosystem

Peraton, a Veritas Capital company, is committed to supporting the nation's most important missions. Across the Department of Health and Human Services (HHS), we provide services to agencies such as: The Centers for Medicare and Medicaid Services (CMS), Food and Drug Administration (FDA), Centers for Disease Control (CDC), Office of Inspector General (OIG), and National Institutes of Health (NIH). Peraton also is a trusted advisor to the nation's defense healthcare ecosystem, providing critical support to the Department of Defense (DoD), Defense Manpower Data Center (DMDC), Defense Health Agency (DHA), Defense Information Systems Agency (DISA), and additionally to the Social Security Administration (SSA).

Veritas Capital has made significant investments in technologies to support the healthcare eco-system and the federal government. In addition to Peraton, Veritas Capital healthcare companies, depicted in **Figure 1**, can be utilized for solutions to the Federal Government.

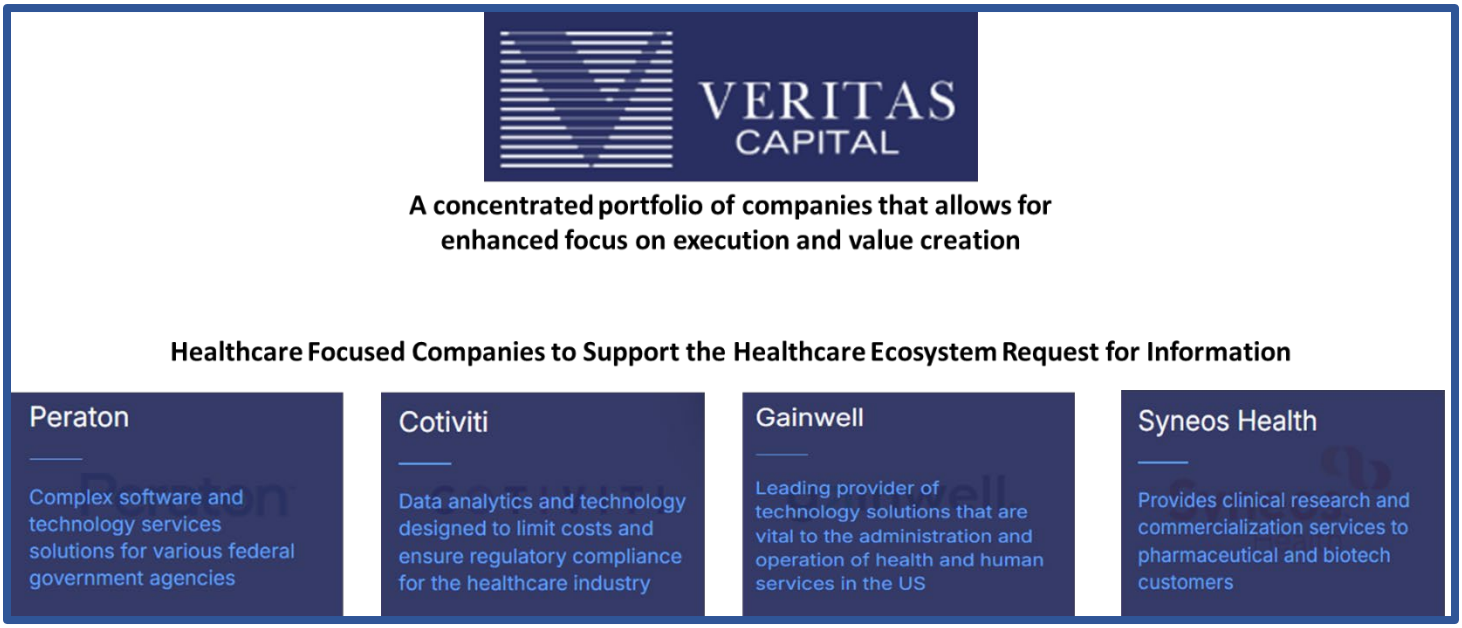

Figure 1: *Veritas Capital provides focused delivery by leveraging expertise across the power of healthcare platforms.*

Peraton's response to this Request for Information reflects our commitment to shaping the future of healthcare technology alongside CMS. With decades of experience supporting complex federal systems—including CMS, DoD, and SSA—we deliver secure, scalable, and user-centered digital solutions tailored to the needs of programs like Medicare and Medicaid. Our feedback emphasizes practical use cases, public-private collaboration, and innovations aligned with CMS's mission to improve health outcomes for all Americans, reduce disparities, and enhance beneficiary experience.

Peraton has deep expertise in modernizing systems that support Medicare claims processing, Medicaid eligibility and enrollment, and value-based care models. Our approach focuses on enterprise interoperability—leveraging HL7 FHIR and other standards to connect CMS, state agencies, providers, and beneficiaries in real time—while ensuring robust cybersecurity and compliance with HIPAA, FedRAMP, and MARS-E frameworks.

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Peraton prioritizes equity and accessibility, designing comprehensive, multilingual, 508-compliant tools that meet the complex needs of Medicare and Medicaid populations, including dual-eligibles and underserved communities. We also responsibly integrate AI, machine learning, and automation to reduce administrative burden, detect fraud, and close care gaps, supporting CMS's transition toward proactive, data-driven oversight.

Demonstratable solutions that Peraton can use to enable the success of the RFI include but are not limited to:

- Secure Digital Signature Technology
- Flexible, Secure Single-Sign-On (SSO) Enablement for large diverse sets of users
- Point of Care products and services to empower providers and patients with real-time data for their Electronic Healthcare Record
- Provide and exchange data from the source connections with APIs and FHIR transactions
- Digitally enhanced native mobile application technology to empower beneficiaries with real-time data, connect with providers, make payments, and report suspected fraud.

To sustain long-term innovation, we advocate for agile, collaborative models—such as public-private innovation labs and DevSecOps delivery pipelines—alongside cloud-native, cost-effective architectures. In short, Peraton offers a mission-aligned, future-ready approach to modernizing Medicare and Medicaid for improved care delivery and better outcomes for all Americans.

We look forward to further discussions from this RFI to implement solutions that improve outcomes for CMS, beneficiaries, and stakeholders.

Omer Khan
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2.0 Response to RFI Questions

B. Patients and Caregivers

1. *Patient Needs*

PC-1. What health management or care navigation apps would help you understand and manage your (or your loved ones') health needs, as well as the actions you should take?

A comprehensive enhanced digital health management platform can significantly improve how patients—and, when necessary, their surrogates—interact with healthcare providers and manage their well-being. These platforms offer numerous benefits, starting with the simplified scheduling and management of medical appointments. Patients can easily book, adjust, and reschedule both in-person and telemedicine visits across various providers, offering crucial flexibility and accessibility. Personalized reminders for appointments, vaccinations, or health screenings also promote adherence to preventive care, tailored to individual needs.

Medication management is another core function. Users can securely reorder prescriptions, receive timely refill notifications, and send requests directly to their providers. Many platforms also incorporate features for checking drug interactions, greatly enhancing patient safety. Additionally, caregivers can leverage these tools to oversee the medication schedules and proactive care of loved ones, simplifying family-based care coordination.

The platforms also facilitate direct communication with healthcare providers through secure messaging or email. This allows patients to ask questions, receive medical advice, or follow up on treatment plans without needing an in-person visit. They also support the organization of health data, such as symptoms and questions, to make provider interactions more efficient. Patients can track their health trends using data from wearable devices—like heart rate, sleep patterns, or glucose levels—and share this information with their doctors to proactively identify and address potential issues.

Access to medical information is a critical feature. Patients can promptly view lab and radiology results, often with plain-language explanations to foster better understanding. The ability to track changes in results over time empowers patients to engage more actively in their health management. Such platforms also help interpret complex medical information, providing clear summaries and suggestions for next steps. Furthermore, they send proactive health alerts, such as annual checkup reminders or preventive screening recommendations, based on a patient's age, medical history, and risk factors.

Finally, care coordination is vastly improved through centralized platforms that share and update medical records across all involved providers. This interoperability ensures that every member of a patient's care team has access to the most current information, reducing errors and enhancing the continuity of care. In essence, integrated health platforms not only streamline daily health tasks but also foster stronger patient engagement, improve communication with providers, and lead to more proactive, personalized healthcare experiences.

PC-2. Do you have easy access to your own and all your loved ones' health information in one location (for example, in a single patient portal or another software system)?

Patient portals can offer significant benefits, including instantaneous access to critical health information, which enhances patient engagement and care coordination. These platforms provide 24/7 access to lab results, radiology reports, appointment schedules, and provider contact information, empowering patients to manage chronic conditions effectively. Secure messaging improves communication with healthcare providers, reducing unnecessary visits (Dendere et al., 2019; Kruse et al., 2018). What is more, portals streamline administrative tasks like scheduling and bill payments, with 98% of users accessing test results in 2019 (Bizmatics, 2024). Caregivers also benefit by monitoring loved ones' health with authorized access, improving care coordination for family members.

A key challenge is the fragmentation caused by multiple provider-specific portals. Each healthcare provider, health system, or service often operates its own distinct portal, creating inconvenience for patients and caregivers who must navigate separate platforms to access comprehensive health information. For example, a

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patient can access their own personal and immediate family members' portals, but managing multiple logins and interfaces is cumbersome. This lack of integration reduces efficiency and user satisfaction. In the future, greater value could be achieved by streamlining patient portals into a unified system, allowing seamless access to all health data across providers and services, which would greatly enhance the beneficiary experience and care coordination.

PC-3. Are you aware of health management, care navigation, or personal health record apps that would be useful to Medicare beneficiaries and their caregivers?

Understanding the Medicare eco-system with the technical infrastructure, how and where the beneficiary's data is at any point in the lifecycle, and the privacy and security aspects to access to the sensitive data, is a key component in enabling any technologies that will allow for access to information. There are many aspects to the Medicare beneficiary eco-system with connections to hundreds of internal and external systems. Beneficiaries also may have multiple aspects of care outside of the Medicare care system – such as, Medicaid, Veteran's health care, Supplemental Insurance, dental care, prescription drug care, and others.

Coordination for care navigation, and access to personal health records will need to be carefully coordinated with current experts in the systems – with consideration to single sign-on access, security threats from bad actors, data governance and management, and the expense of storage for these records of size and magnitude. If a new care navigation system and/or health record application were to be implemented, experts in the current environment and policies will need to be engaged and involved in the application development. In addition, understanding the end-user, with human-centered-design and User Experience (UX) integrated in design, development and testing.

Without partnerships in government and industry experts, the government could be at risk for disruptions in care, unnecessarily expensive and costly implementations, and the potential for bad actors to have increased access to an unsecured environment.

Industry partners with deep domain knowledge are able to add the value needed to add the functionality and enhance the systems to enable better care management and access to data for beneficiaries and their caregivers. Trusted partners, such as Peraton, have applications with advanced technologies in like-systems in other agencies – such as the Military Health System, Veteran's Health Administration, Veteran's Benefit Administration, and Defense Health Agency – that can be leveraged for the goals across the Administration.

Demonstratable examples include:

1. Leveraging the UX-designed mobile applications Peraton created for SSA, Army Recruiting Command, U.S. Military Academy WestPoint, First Responders, and MyFamily Health Portrait CDC Cancer detection. A mobile application that integrates with Medicare beneficiary claims data, enrollment data, provider data, point of care information, payment data, and reporting of fraud, waste, and abuse in real-time, will provide the beneficiary and their caregivers information at the tip of their hand.
2. Leveraging to experience from the continuum of care and connection to Electronic Health Records (EHR) that is used with the Joint Operational Medicine Information Systems (JOMIS) with Defense Healthcare Management Systems (DHMS) for the Department of Defense EHRs. The JOMIS ecosystem is a network of interconnected software applications, tools, services and platforms that work together to create a cohesive environment for end users. With core capabilities (e.g. point of injury, triage), advanced capabilities (e.g. medication dosing guidelines, patient risk scores, clinical data worksheets), ancillary services (e.g. pharmacy, dental, radiology), and data movement (e.g. data seeding, computable data records, devise to user ratio), there is seamless integration, holistic workflows, enhanced collaboration, scalability, centralized data management, innovation, and cost efficiencies.

   In addition, working with Peraton partners, the OpMed application enables medical personnel to facilitate the efficient and accurate documentation at point of care that could be used in the HHS community care centers today for Medicare beneficiaries. The application provides enhanced communication among medical personnel and ensures timely access to patient records, supporting effective triage and treatment with full off-line capabilities for rural areas that many of the HHS facilities reside in.

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PC-4. What features are missing from apps you use or that you are aware of today?

In reviewing the MyMedicare.gov online application from the UX perspective – both in the online application, and through the mobile application – we found that there were design issues, information not connected, and general issues of confusion for the beneficiary and their caregiver that can easily be addressed using information and workflows available today from existing contractors that develop and manage the systems that they pull from. A few examples are noted below:

1. **Creating an account** – the beneficiary has to enter the Medicare number and start date; the Medicare ID number has letters/numbers with dashes - but the system only allows for uppercase and no dashes - it is confusing to unsavvy users to create an account because the example and the ID card both had the dashes and if you're on a phone - that's hard to input the all uppercase

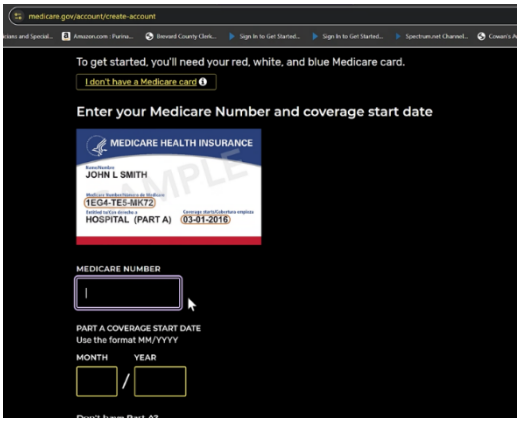

Figure 2: *Medicare.gov application was created without the user in mind.*

2. **Creating a username and password** was excessive and is hard for a Medicare beneficiary to remember (**Figure 3**). A single-sign-on (SSO) feature will be much easier to use (see section PR-9).

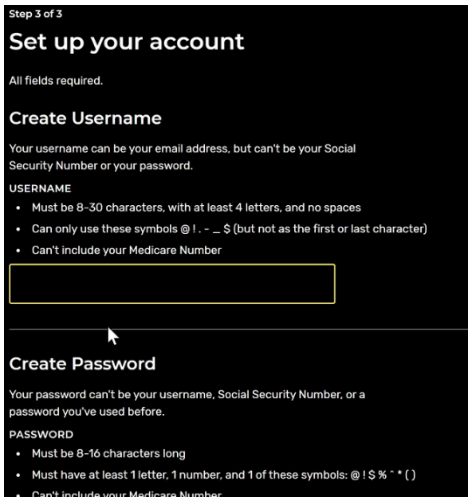

Figure 3: *Medicare.gov without SSO capabilities is hard to navigate for users.*

3. **Overlaid screens on website** - the "popups" overlaid other screens - for example when clicking on a button to move to another part of the application, it kept the home screen in the background too.

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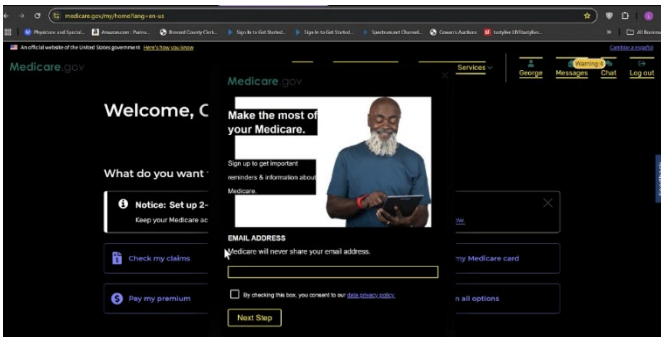

Figure 4: *Medicare.gov overlays screens; native mobile applications can be better built for ease of use .*

4. Finding a **return to "home screen" button** was confusing – for the Medicare beneficiary to get back to the home screen, the user has to hit their name "button" on the top right (see **Figure 5** below) and then click "My Account" to get to the home screen or main menu.

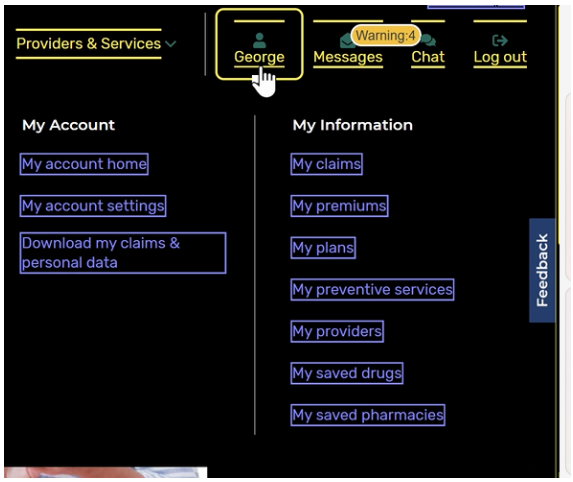

Figure 5: *Medicare.gov lacks intuitive navigation features.*

a. What apps should exist but do not yet? Why do you believe they do not exist yet?

There is no current "Medicare Mobile Application" (MMA) available to download on your phone. The user must login from their phone to the website, which is a replication of the online web application. Empowering beneficiaries to use a native Mobile Medicare Application developed by organizations that are already familiar with the user experience (like the prototype created by Peraton below in **Figure 6**), will provide beneficiary mobile ease of use with a single-sign-on, enrollment, digital ID, claims, provider geo location look-ups, and real-time fraud prevention – all using the source data securely through available APIs in the current systems.

The key to success is involving the UX in the design. Leveraging rapid prototyping to develop clickable Proof of Concepts (PoCs) to validate UX with mock data, Peraton can iterate to connect to the systems and data we already support to develop an MVP working demo quickly. Peraton's current stewardship of CMS' mission systems for Beneficiary, Claims, and Fraud, Waste, and Abuse prevention provide direct access to the data, APIs, and functionality to enable a full featured mobile application for beneficiaries.

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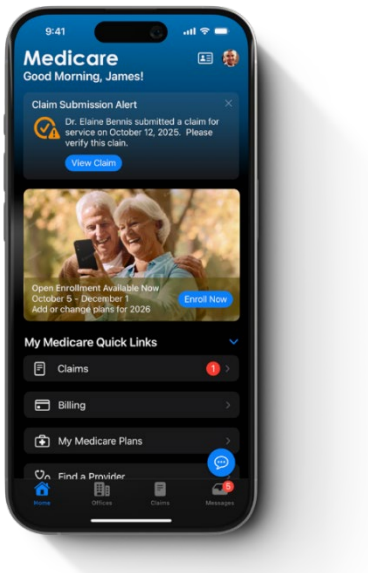

Figure 6: *Peraton's native Medicare mobile application prototype incorporates all of the functionality and data for beneficiaries to seamlessly and securely navigate a complex healthcare system – including real-time potential fraud alerts.*

b. What set of workflows do you believe CMS is uniquely positioned to offer?

Similar to the application and system development provided to other agencies mentioned above, Peraton and CMS experts are uniquely positioned to use the lessons learned and the knowledge of the eco-system, environment, and data at CMS, to create a nation-wide health eco-system for patients at large. Having a partner that can work with the patient information, no matter where they are in their journey, without disrupting their care or their privacy to information, is critical to the success of the changes needed on this level of scale.

In addition, the use of the HHS providers in their facilities, to prove out the workflows, point of care applications and system changes will be critical to the success and adoption of changes needed to change the flow/process of information. Using HHS providers, and those that do work with CMS, you can use a specific system to ensure that data is shared, information is standardized, and policies are followed from the providers to the payer systems.

PC-5. What can CMS and its partners do to encourage patient and caregiver interest in these digital health products?

Establishing a Voluntary Validation Framework for Digital Health Tools

CMS is uniquely positioned to spearhead the development of a voluntary, evidence-based validation framework for digital health technologies. Although not a regulatory body like the FDA, CMS's role as a standards setter enables it to bring together key stakeholders—including patients, providers, and payers—to foster a more transparent and trustworthy digital health ecosystem. In partnership with agencies such as the National Institutes of Health (NIH) and the Agency for Healthcare Research and Quality (AHRQ), CMS can establish clinical and outcomes-focused benchmarks, provide quality assurance guidance in collaboration with accrediting bodies, and develop secure validation environments using de-identified CMS data. These initiatives would promote scientifically validated, user-centered digital tools while preventing the proliferation of unverified products.

Criteria for Digital Health Product Evaluation

To establish a transparent and scalable review process for digital health tools, CMS can implement standardized evaluation criteria across several critical areas. First, clinical validity must be demonstrated,

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requiring tools to prove effectiveness through rigorous randomized controlled trials or validated real-world evidence. Second, usability and accessibility are paramount; the design must effectively cater to all Medicare and Medicaid beneficiaries, including individuals with disabilities. Third, data privacy is non-negotiable, demanding strict adherence to HIPAA and 42 CFR Part 2 regulations to protect sensitive health information. Fourth, a tool's health equity impact should be assessed, focusing on its ability to reduce healthcare disparities or address social determinants of health. Fifth, interoperability is vital, necessitating compliance with HL7 FHIR standards for seamless data exchange across the healthcare ecosystem. Finally, cost-effectiveness is essential, with tools needing to demonstrate the capacity to lower overall healthcare costs without compromising care quality.

Enabling Adoption by Patients and Caregivers

Achieving widespread adoption of digital health tools within Medicare and Medicaid populations requires a multifaceted strategy. This includes expanding APIs and patient portals to simplify access to personal health data, empowering beneficiaries to manage their information. It's also crucial to deploy Digital Health Navigators in community-based aging and disability centers, offering hands-on support and training to bridge the digital literacy gap. Furthermore, CMS can support robust reimbursement for remote care services, including innovative AI-enabled tools, to incentivize providers. Concurrently, comprehensive outreach campaigns are necessary to build trust and enhance overall healthcare literacy among beneficiaries. Finally, addressing fundamental connectivity gaps through device and internet subsidies, potentially via Medicaid waivers and Medicare Advantage plans, will ensure equitable access for all.

Challenges in Digital Health Implementation

Despite their promise, innovative digital health applications for physician workflows (e.g., quality reporting, documentation, billing) face significant hurdles. These common obstacles include high financial costs due to development and implementation expenses, uncertain return on investment, and inadequate reimbursement models, especially for smaller or rural practices often lacking essential broadband infrastructure. Workflow integration challenges and the burden of staff training disrupt established clinical routines, leading to user resistance and increased administrative load. Pervasive interoperability issues and data silos continue to impede seamless data exchange, fragmenting patient information and hindering comprehensive care coordination. Furthermore, legitimate privacy and security concerns surrounding sensitive health data, alongside the complexities of achieving and maintaining HIPAA compliance, add layers of risk and regulatory burden. Issues like physician burnout, often exacerbated by the administrative demands of existing digital tools, and a broader digital divide among patients with varying technological literacy and access, collectively limit the effective adoption and full utilization of these promising innovations.

PC-6. What features are most important to make digital health products accessible and easy to use for Medicare beneficiaries and caregivers, particularly those with limited prior experience using digital tools and services?

There is no Medicare application that is a one-stop shop. While there is mymedicare.gov, it is a replication of the web version that is accessible to the beneficiaries and caregivers. See section PC-4 for examples to ease of use for Medicare beneficiaries.

PC-7. If CMS were to collect real-world data on digital health products' impact on health outcomes and related costs once they are released into the market, what would be the best means of doing so?

CMS is strategically positioned to boost physician interoperability by embedding standardized data formats, automation, and incentives into existing clinical workflows, all while minimizing administrative burden. By aligning interoperability efforts with physicians' current motivations, such as quality reporting and care coordination, CMS can facilitate seamless, high-quality data sharing that truly benefits patient care.

CMS can drive interoperability through a multitude of key strategies:

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- **Mandating Standards and APIs**: Enforcing the use of Fast Healthcare Interoperability Resources (FHIR) and APIs (HL7 International, n.d.) ensures fluid data exchange. This can be supported by enforcing EHR certification through the ONC Health IT Certification Program and providing funding for API implementation (Office of the National Coordinator for Health IT, n.d.).
- **Embedding Tools within EHRs**: Integrating interoperability tools directly into Electronic Health Records (EHRs) can automate data sharing during routine clinical tasks like charting or discharge. CMS can update certification criteria and link these efforts to Merit-based Incentive Payment System (MIPS) Promoting Interoperability measures (Centers for Medicare & Medicaid Services, n.d.).
- **Incentivizing Data Exchange:** CMS can offer incentives for interoperability through value-based programs like MIPS and Alternative Payment Models (APMs), rewarding metrics such as API-based exchanges and funding training for Health Information Exchanges (HIEs) or the Trusted Exchange Framework and Common Agreement (TEFCA) (Centers for Medicare & Medicaid Services, n.d.; Adler-Milstein & Jha, 2017).
- **Empowering Patients:** Enabling beneficiaries to share their health data via patient portals or apps, as mandated by the 21st Century Cures Act (2016), can significantly reduce physician workload by allowing patients to proactively manage and provide their information. CMS can further incentivize education around portal use.

To effectively limit physician burden, CMS can streamline workflows by integrating single-click HIE sharing and AI-driven data pre-population directly into EHRs. Aligning interoperability data with existing quality reporting requirements, such as MIPS and HEDIS measures (National Committee for Quality Assurance, n.d.), eliminates redundant documentation. CMS can also fund non-physician task delegation through programs designed to transform clinical practice models (Kruse et al., 2018).

CMS can make interoperability intuitive by leveraging existing workflows. Reusing data points already collected for MIPS or HEDIS, such as Continuity of Care Documents (CCDs), aligns with physicians' reimbursement goals and promotes efficient care coordination (Holmgren & Adler-Milstein, 2021). Automating referral and discharge data sharing (e.g., sending lab and radiology results to specialists via FHIR) improves patient outcomes within value-based care models. Linking patient portal use to interoperability metrics taps into physicians' existing MIPS tracking efforts. Furthermore, sharing chronic care data, such as A1c and Urine Albumin Creatine Ratio results, via HIEs enhances chronic disease management for conditions like diabetes and kidney disease, ultimately improving patient outcomes (Menachemi et al., 2018).

CMS can drive interoperability by standardizing formats, automating workflows, offering incentives, and empowering beneficiaries. This can be achieved while reducing burden through streamlined processes, decreased documentation, staff support, and clear vendor accountability. Integrating metrics like API exchange success rates into MIPS directly aligns interoperability with physicians' existing motivations for seamless, high-quality data sharing.

PC-8. In your experience, what health data is readily available and valuable to patients or their caregivers or both?

The medical professionals at Peraton which consist of doctors, nurses and technicians consistently affirm that readily accessible health data has proven exceptionally valuable for patients, their caregivers, and family members alike. This accessibility profoundly empowers individuals, enabling them to make more informed decisions regarding their health and care.

This readily available data included crucial information like lab and radiology results, which is vital for understanding diagnoses and assessing treatment effectiveness. An up-to-date list of prescribed medications, complete with dosage and frequency, is invaluable for managing medications, ensuring compliance, and

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safety. Appointment reminders and a history of past appointments were also easily accessible, which proved very helpful.

Beyond these, a clear problem list with associated diagnoses, immunization history, and a record of allergies and adverse reactions are critical for a comprehensive understanding of an overall health status. Access to vital signs over time for hypertension management, summaries of clinic visits, and even discharge notes offered valuable insights and reinforced care instructions. Finally, information regarding referrals streamlined the coordination of a patients care.

All of this accessible data fostered informed decision-making, improved medication adherence and safety, facilitated better communication with healthcare providers, improved continuity of care, and ultimately reduced fears by providing a clear picture of a beneficiary's health journey.

a. **What data is valuable, but hard for patients and caregivers, or app developers and other technical vendors, to access for appropriate and valuable use (for example, claims data, clinical data, encounter notes, operative reports, appointment schedules, prices)?**

   While patients and caregivers are getting better access to some health data, significant challenges remain for truly valuable use. It's tough to get comprehensive clinical data like detailed encounter notes and operative reports, as this information is often unstructured and hard to interpret for most patients or integrate. Claims data is another large obstacle; it's controlled by insurers and not easily accessible, making it difficult for patients to see truly their full healthcare utilization and costs. Getting accurate, personalized prices for healthcare services before treatment is also a major hurdle due to varying negotiated rates. Moreover, managing appointment schedules across multiple different health systems in one place is nearly impossible due to a lack of interconnectedness. Finally, accessing a complete, unified health record for patients who've received care from various providers, and integrating social determinants of health data, continues to be a struggle. These difficulties largely stem from technical issues, privacy concerns, business models that can limit data sharing, and a lack of consistent data standards across the healthcare industry also play into the equation.

b. **What are specific sources, other than claims and clinical data, that would be of highest value, and why?**

   Wearable technologies profoundly benefit patients, doctors, and providers by enabling continuous, proactive health monitoring. For patients, they foster self-management and early issue detection, like fall detection on smartwatches improving elderly safety. For providers, wearables deliver objective, real-time data for enhanced diagnostics, personalized treatment, and remote management of chronic conditions, potentially reducing in-person visits and emergencies. Conditions such as cardiovascular diseases (e.g., continuous ECGs for arrhythmia detection), diabetes (Continuous Glucose Monitors for real-time blood sugar management), sleep disorders, epilepsy (seizure detection alerts), and post-surgical recovery (vital sign monitoring) are greatly improved through this constant oversight, leading to better patient outcomes and peace of mind.

Envision a clinical application where a patient, 1) has an emergency onset of heart palpitations and is seen at an Emergency Room, employs a point of care device that seamlessly and quickly incorporates the clinical data for the immediate transmission to an EHR; or 2) is previously diagnosed with irregular heartbeats, employs a smartwatch incorporating an ECG functionality to facilitate proactive self-monitoring and potential early detection of significant cardiac events. How Wearable and Point of Care Data is Transmitted to an EHR (Using a Smartwatch, Point of Care Provider Devices, and FHIR):

1. **Data Collection**: The provider application similar to OpMed used in JOMIS discussed in section PC-3 collects the patient's data in real-time quickly and easily. The smartwatch continuously monitors the patient's heart rate and periodically performs an ECG recording (e.g., when the patient initiates it or if it detects an anomaly). This raw data (heart rate, ECG waveform, time stamps) is collected by the device.

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2. **Transmission to a Companion App/Cloud**: The OpMed application and device can work offline and also upload the data securely to any EHR with APIs (see Section PC-3). The smartwatch typically syncs this data wirelessly (e.g., via Bluetooth) to a companion mobile application on the patient's smartphone. This app then uploads the data securely to a cloud-based platform or the wearable manufacturer's server. This can be easily integrated into the native mobile application prototyped by Peraton discussed in section PC-4.
3. **Standardization and API (FHIR)**: This is where interoperability comes in. The cloud platform or a third-party data aggregator (like Validic or Human API) uses APIs (Application Programming Interfaces) to extract this raw data. Crucially, they then standardize it into a common healthcare data format, most commonly HL7 FHIR (Fast Healthcare Interoperability Resources). For instance, a heart rate reading would be mapped to a FHIR "Observation" resource with specific codes for heart rate, unit of measure, and the timestamp. An ECG recording might be represented as a FHIR "DiagnosticReport" or "Observation" containing the waveform data.
4. **EHR Integration:** The standardized FHIR data is then sent from the cloud platform to the hospital or clinic's Electronic Health Record (EHR) system (e.g., Epic, Cerner) using FHIR APIs. The EHR's interface receives these FHIR resources and ingests them into the patient's digital chart.
5. **Provider Access:** Within the EHR, the cardiologist can now access a dashboard or a specific section of the patient's chart that displays the continuous heart rate trends, recent ECGs, and any alerts generated by the wearable. This allows the doctor to monitor the patient's condition remotely, identify patterns, and make informed decisions about medication adjustments or further diagnostic tests, all without the patient needing to visit the clinic.

This streamlined process, significantly facilitated by interoperability standards such as FHIR, ensures the seamless integration of valuable real-time physiological data from wearable devices directly into the formal medical record. This capability effectively bridges the gap between a patient's daily life and their clinical care, enabling a more comprehensive and continuous understanding of their health.

c. What specific opportunities and challenges exist to improve accessibility, interoperability and integration of clinical data from different sources to enable more meaningful clinical research and generation of actionable evidence?

To establish a transparent and scalable review process for digital health tools, CMS can implement standardized evaluation criteria across several critical areas. First, clinical validity must be demonstrated, requiring tools to prove effectiveness through rigorous randomized controlled trials or validated real-world evidence. Second, usability and accessibility are paramount; the design must effectively cater to all Medicare and Medicaid beneficiaries, including individuals with disabilities. Third, data privacy is non-negotiable, demanding strict adherence to HIPAA and 42 CFR Part 2 regulations to protect sensitive health information. Fourth, a tool's health equity impact should be assessed, focusing on its ability to reduce healthcare disparities or address social determinants of health. Fifth, interoperability is vital, necessitating compliance with HL7 FHIR standards for seamless data exchange across the healthcare ecosystem. Finally, cost-effectiveness is essential, with tools needing to demonstrate the capacity to lower overall healthcare costs without compromising care quality.

PC-9. Given that the Blue Button 2.0 API only includes basic patient demographic, Medicare coverage, and claims data (Part A, B, D), what additional CMS data sources do developers view as most valuable for inclusion in the API to enable more useful digital products for patients and caretakers?

While Blue Button 2.0 APIs provide crucial information, it does not support many use cases and is informational in nature. Peraton has developed similar robust API frameworks with the Beneficiary Information in the Cloud (BIC) that provides similar data to internal CMS users and external plans to allow for faster dissemination of data in real-time. Additional use cases, like supporting an application for Beneficiaries to

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empower them with their health information, allow providers to share comprehensive health records with the patient and other providers, could not be enabled by these APIs alone. Additionally, with the focus on Fraud, Waste, and Abuse introducing a way for Beneficiaries to notify CMS if they see activity in their claims that they do not recognize could directly tie into CMS' Fraud Prevention System (FPS). Peraton has developed and innovated the FPS system and Fee for Service Part A and Part B claims systems since inception and can implement functionality to notify beneficiaries of activity prior to the claim being paid and integrate them into process similar to the commercial card payments model. The process flows can easily be changed to accommodate this flow, while incorporating APIs to the mobile application and the investigators to review in real time.

a. **What difficulties are there in accessing or utilizing these data sources today?**

   Without detailed health record data the use cases around the existing Blue Button 2.0 API are limited.

b. **What suggestions do you have to improve the Blue Button 2.0 API experience?**

   CMS has moved in the right direction with an API-First posture in recent modernization efforts. Several APIs have been developed at CMS, including the BIC API noted above. Ideally, CMS should leverage a suite to manage their APIs to determine if any of the internal system to system APIs can be extensible to external uses like the Beneficiary (via a mobile app, for example). Another factor is the data sources for the APIs are disparate within CMS. The data that is leveraged most across CMS, is the claims data. An effort to consolidate the Fee-For-Service (FFS) claims processing systems into a singular real-time payment system will enable a one stop shop for the majority of the data required across CMS.

c. **Is there non-CMS data that should be included in the API?**

   As noted in section PC-3, providers can get real-time patient data in the hands of the patient at point of care using the OpMed application and tool, and providing an API not only to the EHR of record for the provider, but also directly to the patient's mobile application (discussed in section PC-4).

PC-11. How are health information exchanges (HIEs) currently helping to advance patient access to health information in the real world?

Health Information Exchanges (HIEs) are fundamentally reshaping patient access to health information by enabling the secure, electronic sharing of medical data among diverse healthcare providers and directly with patients. Through HIEs, individuals gain immediate access to their comprehensive health records via patient portals, empowering them to actively manage their care, verify information, and engage in shared decision-making. For providers, HIEs deliver a holistic patient view, significantly enhancing care coordination, reducing redundant tests and medical errors, and ensuring seamless transitions across different healthcare settings. This streamlined information flow not only improves patient safety and outcomes but also boosts efficiency by reducing administrative burdens and lowering healthcare costs. Furthermore, HIEs are vital for public health initiatives, facilitating disease surveillance and reporting. By breaking down data silos between disparate EHR systems, HIEs drive interoperability, ensuring compliance with evolving regulations like the Information Blocking rule, and ultimately fostering a more connected, efficient, and patient-centered healthcare ecosystem.

a. **How valuable, available, and accurate do you find the data they share to be?**

   The data shared through HIEs offers substantial value by providing a comprehensive patient view, which is critical for enhancing care coordination, reducing errors, and empowering patients in their health management, all while supporting public health. Data availability has significantly increased due to growing EHR adoption and regulatory mandates, with HIEs offering various exchange methods, though universal access for every patient from every encounter is still evolving. Regarding accuracy, HIE data is generally good and improving, but its quality is ultimately reliant on the accuracy of source EHRs, effective data standardization, robust patient matching to avoid incomplete or mis-linked records, and timely updates. Despite these inherent challenges, the benefits of HIEs in delivering more complete and generally accurate information for improved healthcare outcomes far outweigh their current limitations.

b. **Are there particular examples of high-performing HIE models that you believe should be propagated across markets?**

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Increasing the successful numbers of HIE models requires focusing on those that are technically agile, well-governed, and demonstrate clear value beyond basic data exchange. Key characteristics include hybrid architectures actively embracing FHIR standards for flexible, real-time data flow, as seen in leaders like MiHIN and IHIE (MiHIN, nd). Strong, collaborative governance and sustainable funding models are crucial, often involving diverse stakeholders and innovative fee structures. High-performing HIEs prioritize high-value use cases like real-time encounter notifications, population health management, and automated public health reporting, while also ensuring information blocking compliance and direct patient access. Finally, their scalability and participation in broader national networks like Carequality are essential for achieving truly widespread interoperability, allowing access to patient data across geographical boundaries (Carequality, nd).

c. **What is the ongoing role of HIEs amidst other entities facilitating data exchange and broader frameworks for data exchange (for example, vendor health information networks, TEFCA, private exchange networks, etc.)?**

HIEs maintain a critical and evolving role in the health data exchange landscape by serving as foundational infrastructure with local expertise, bridging interoperability gaps between diverse EHR systems, and actively adapting to broader frameworks like TEFCA by becoming or participating within QHINs. They are central to driving the adoption of modern standards like FHIR, enabling advanced use cases, and moving beyond basic data exchange to offer valuable services such as population health management, analytics, and robust patient matching. Essentially, HIEs remain indispensable local and regional connectors, aligning with national initiatives while providing essential value-added services that enhance care coordination and public health. HIEs remain pivotal in the evolving health data exchange landscape, serving as a foundational infrastructure with vital local expertise and established trust networks. They bridge the gap between diverse Electronic Health Record systems, facilitating interoperability by normalizing data across various standards, from legacy HL7 to modern FHIR. Crucially, HIEs are adapting to broader national frameworks like TEFCA, often becoming Qualified Health Information Networks (QHINs) or integral participants, thereby extending nationwide exchange to local providers and patients. Beyond basic data transfer, HIEs are increasingly offering value-added services such as population health management, advanced analytics, and granular consent management, cementing their role as essential regional connectors that align with evolving regulations and drive the future of a more connected and patient-centered healthcare system.

PC-12. What are the most valuable operational health data use cases for patients and caregivers that, if addressed, would create more efficient care navigation or eliminate barriers to competition among providers or both?

Addressing key operational health data challenges for patients and caregivers is pivotal for optimizing care navigation, overcoming barriers to cost-effective and value-based care, and profoundly benefiting American citizens. The primary obstacle remains the fragmentation of health data and the absence of seamless interoperability throughout the healthcare ecosystem. Effective implementation of the following data use cases would yield significant positive impacts:

- **Comprehensive, Accessible Patient and Caregiver Health Records**: This use case involves providing patients and their designated caregivers with immediate, secure, and easily understandable access to all health information—including historical data, current medications, lab results, imaging, specialist notes, and care plans, as well as data from wearable devices—via a centralized patient portal or personal health record (PHR) system. This fosters efficient care navigation by reducing redundant tests and empowering patients/caregivers in decision-making, while promoting cost-effective care by identifying care gaps and enabling informed choices. It supports value-based care through shared decision-making and holistic patient views, significantly reduces caregiver burden by simplifying coordination and medication management and enhances patient safety by minimizing errors.
- **Real-Time Data Integration from Wearables and Remote Monitoring Devices**: This involves ensuring a continuous, seamless flow of real-time physiological data, such as heart rate, blood

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pressure, glucose levels, activity, and sleep patterns, from patient-owned wearables and remote patient monitoring (RPM) equipment directly into the formal medical record for care team access. This facilitates efficient care navigation through proactive intervention for chronic conditions and early detection of deterioration, thereby reducing emergency visits and hospitalizations. It drives cost-effective care by shifting from reactive to preventive approaches and supports value-based care with continuous monitoring and personalized plans, ultimately improving chronic disease management and population health outcomes. Furthermore, it empowers patients through active health tracking and self-management.

- **Predictive Analytics for Proactive Risk Identification and Intervention**: This use case leverages AI and machine learning to analyze aggregated patient data, including clinical, social determinants of health (SDOH), and behavioral data, to predict deterioration, identify individuals at high risk for hospital readmissions or disease progression, and forecast potential care gaps. This enhances efficient care navigation by enabling timely interventions and optimizing resource allocation for high-risk patients. It achieves cost-effective care by mitigating the financial burden of late-stage complications and advances value-based care through preventive strategies and improved population health. Crucially, it promotes health equity by identifying and addressing SDOH factors in vulnerable populations.
- **Streamlined Appointment Scheduling and Care Coordination**: This involves implementing digital platforms for patients and caregivers to easily schedule, reschedule, and manage appointments across various providers and health systems, integrating communication tools for seamless care team coordination. This improves efficient care navigation by eliminating administrative burdens and enhancing access to care, while fostering cost-effective care by optimizing provider schedules and reducing missed appointments. It also elevates the patient experience through increased convenience and satisfaction.
- I**ntelligent Medication Management and Adherence Tools**: This final use case involves deploying digital tools, such as smart pill dispensers, mobile reminder apps, and telehealth counseling, to assist patients and caregivers in managing complex medication regimens, tracking adherence, and providing real-time support for missed doses. This boosts efficient care navigation by reducing medication errors and enhancing therapeutic outcomes. It also contributes to cost-effective care by preventing non-adherence-related emergencies and directly supports value-based care by ensuring optimal treatment efficacy.

a. Examples may include the following:

Envision a healthcare ecosystem where patients and their caregivers are empowered with confidence and a profound sense of control over their medical journeys. This transformative vision is attainable through the strategic and intelligent application of operational health data, which will drive unprecedented efficiencies in care delivery and will foster genuine, value-based competition among healthcare providers.

Binding cost estimates for pre-defined periods. Instead of today's confusing costs, imagine receiving a guaranteed, binding cost estimate for any procedure or treatment, covering all anticipated charges for a set period. This isn't just a number; it's financial peace of mind, empowering patients to make informed decisions without fear of hidden fees. This transparency would also encourage providers to offer clear, competitive pricing.

Viewing provider schedule availability. Envision real-time, comprehensive visibility into provider schedules across entire networks. This means intuitive platforms showing available slots, next-day appointments, and waitlist options. This seamless access eliminates scheduling bottlenecks, allowing instant bookings at your convenience. Such transparency would also drive competition, as providers with efficient scheduling would naturally attract more patients.

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Using third-party apps for appointment management. Imagine managing your healthcare journey through interoperable, secure third-party apps of your choosing. These apps would integrate with all your providers, offering a holistic view of appointments, pre-visit instructions, and medication reminders. This empowers patients with personalized, proactive tools that fit their lifestyle, breaking down data silos and fostering innovation in patient-centric solutions.

Accessing patient-facing quality metrics. Picture a world where easily consumable, patient-facing quality metrics are readily available for every provider. This goes beyond simple ratings, offering clear data on outcomes, patient satisfaction, and wait times. Greater transparency empowers patients to make truly informed decisions about their care and ignites powerful competition among providers to continuously improve quality.

Finding the right provider for specific healthcare needs. Imagine a cutting-edge, AI-driven platform that serves as your dedicated healthcare guide, precisely aligning you with the most suitable provider for your unique health requirements. This intelligent system would factor in your complete medical history, personal preferences, insurance coverage, and comprehensive patient feedback to offer tailored recommendations. It would effectively remove the uncertainty from selecting a healthcare professional, guaranteeing that every individual is connected with optimal care. Concurrently, this platform would enable providers to efficiently reach patients who genuinely benefit from their specialized services.

Achieving these advancements would transcend mere operational efficiency, fostering a healthcare environment where patients are empowered, caregivers receive vital support, and providers are driven by healthy competition to deliver the most transparent, accessible, and highest-quality care possible.

b. What use cases are possible today?

All five of the operational health data use cases described are possible and actively being implemented today. This capability stems from significant advancements in technology, the increasing adoption of digital health solutions, and a growing recognition of the immense value that data brings to healthcare.

In view of that, comprehensive, accessible patient and caregiver health records are largely a reality, with most healthcare providers offering patient portals that provide secure access to medical histories, medications, lab results, and more. Standards like FHIR are pivotal in enabling data exchange between diverse systems, even if truly unified records across all providers remain an ongoing aspiration. Similarly, real-time data integration from wearables and remote monitoring devices is rapidly expanding. Wearable technology and Remote Patient Monitoring (RPM) programs are now commonplace, allowing physiological data to seamlessly flow into medical records, enabling proactive health management. Furthermore, predictive analytics for proactive risk identification and intervention is an active area of development and deployment. AI and machine learning are being used to analyze vast datasets, forecasting risks like hospital readmissions or disease progression, which allows for timely, targeted interventions. Streamlined appointment scheduling and care coordination are widely available, with online scheduling being a standard feature of patient portals, enhancing access and reducing administrative burdens. Lastly, intelligent medication management and adherence tools are supported by a growing array of solutions, including reminder apps, smart pill dispensers, and telehealth counseling, all designed to improve medication adherence and reduce errors. While these use cases are feasible and in motion, the persistent challenge remains achieving true, seamless interoperability across the entire healthcare ecosystem.

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c. What should be possible in the near future?

In the near future, Peraton envisions widespread adoption and maturation of current healthcare data capabilities. This means:

- **Pervasive Interoperability with Patient Data Ownership**: Expect seamless, system-agnostic data flow, allowing patients to control and direct their complete health records to any provider or app.
- **AI-Powered Personalized Health Assistants**: AI will move beyond basic predictions to offer highly personalized, prescriptive guidance and proactive nudges based on individual health data and preferences.
- **Dynamic, Unified Care Plans**: Care plans will become real-time, collaborative, and adaptable across all providers and settings, ensuring consistent, coordinated care transitions.
- **Automated Administrative Burdens**: Enhanced interoperability and AI will significantly reduce frustrations from prior authorizations, claims processing, and complex scheduling.

d. What would be very valuable but may be very hard to achieve?

Achieving the following systemic transformations, while offering substantial value, presents profound challenges.

- **A Truly Proactive, Preventative Healthcare System:** Transitioning from illness management to population-wide prevention necessitates fundamental reforms in funding models, societal health behaviors, and the comprehensive integration of public health and social services into clinical workflows.
- **Seamless, Trustworthy Global Health Records:** The creation of a single, lifelong, universally accessible, and patient-controlled health record spanning international borders and diverse healthcare systems faces immense hurdles in establishing global interoperability standards, reconciling disparate privacy laws, and ensuring robust security at an unprecedented scale.

2.1.1 Information Blocking and Digital Identity

C. Providers

1. *Digital Health Apps*

PR-1. What can CMS and its partners do to encourage providers, including those in rural areas, to leverage approved (see description in PC-5) digital health products for their patients?

To significantly boost the adoption of approved digital health products, CMS and its partners need a comprehensive approach that addresses key areas. CMS can expand and clarify financial incentives and reimbursement models, ensuring providers are adequately compensated for digital health services like telehealth and remote patient monitoring, aligning these with value-based care initiatives, and offering grants or subsidies to cover upfront costs for rural providers, which can include crucial broadband infrastructure improvements (CMS, 2025). Alongside financial support, robust education, training, and technical assistance programs are essential; these can be tailored to help providers seamlessly integrate digital tools into their existing EHR systems and workflows, while also fostering platforms for sharing best practices and success stories to build confidence (Calcium Health, n.d.). Furthermore, driving interoperability and standardization is critical, requiring efforts to promote seamless data exchange between different platforms using standards like FHIR APIs to reduce data silos, standardize data collection, and work towards unifying patient access across multiple portals to ease administrative burden (CMS, 2025; Dendere et al., 2019). Finally, building awareness and trust by publicizing successful implementations, particularly in rural areas, and clearly communicating stringent data security and privacy measures will be vital to alleviate concerns and encourage broader adoption (AudioEye, 2024; HHS.gov, n.d.).

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a. What are the current obstacles

Several significant obstacles currently hinder widespread digital health adoption among providers. Financial constraints are a major barrier, encompassing high upfront costs for hardware, software, and implementation, coupled with uncertain ROI due to inconsistent reimbursement policies, particularly affecting smaller or rural practices (Calcium Health, n.d.). Rural areas also frequently contend with a lack of reliable, high-speed internet, which is crucial for many digital health applications (CMS, n.d.b). Another challenge lies in workflow integration and the associated training burden; seamlessly integrating new digital health tools with existing electronic health record (EHR) systems often proves difficult, leading to fragmented workflows and increased administrative load (Calcium Health, n.d.; Healthtechprime.com, n.d.). Furthermore, the time-consuming training required for providers and staff can be a deterrent, with some resisting new technologies out of comfort with established practices (Calcium Health, n.d.). Interoperability and data silos significantly impede comprehensive care coordination, as different digital health platforms and EHRs often fail to communicate effectively, resulting in fragmented patient data (AudioEye, 2024; Healthtechprime.com, n.d.). This issue is exacerbated by the prevalence of multiple provider-specific portals, which complicate patient and provider access to holistic health information (Dendere et al., 2019). Privacy and security concerns are also paramount, with legitimate worries among both providers and patients about the security of sensitive health information in digital environments, including the risk of data breaches and the complexities of navigating HIPAA compliance (AudioEye, 2024; American Medical Association, n.d.b). Finally, patient engagement and the digital divide present challenges, as varying levels of digital literacy and access to necessary technology among patients, particularly in rural or underserved areas, can impact their ability to effectively use digital health tools, and sustaining long-term patient engagement can also be difficult (Oxford Academic, 2025; Healthtechprime.com, n.d.).

b. What information should providers share with patients when using digital products in the provision of their care?

When providers recommend digital products as part of patient care, it's important to offer comprehensive and transparent information to ensure informed consent and effective use. Clearly explaining the purpose and benefits of the digital product is paramount, such as how it can enhance self-management, improve communication, or facilitate better care coordination (Samsung Business Insights, 2016). Providers and their teams must also detail how the product works, providing clear instructions on its usage, how data will be collected and reviewed, and how this information will be utilized by the care team. Critically, information regarding data privacy and security must be thoroughly communicated, outlining measures like encryption and HIPAA compliance, and clarifying who will have access to the patient's PHI (HHS.gov, n.d.; American Medical Association, n.d.b). Patients should be fully aware of their responsibilities, such as consistent data entry and maintaining secure login credentials. Furthermore, providers are obligated to discuss any potential limitations, risks, and emergency procedures, emphasizing that digital tools are not substitutes for emergency care and providing clear instructions for when to seek immediate medical attention. In conclusion, patients should be informed about any associated costs and unequivocally understand their right to opt out of using the digital product at any time without compromising their access to traditional care.

c. What responsibilities do providers have when recommending use of a digital product by a patient?

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When recommending digital health products, providers bear substantial responsibilities to ensure patient safety, deliver effective care, and uphold ethical practice. This involves a rigorous process of due diligence and vetting, where providers must verify that the digital product demonstrates evidence of clinical efficacy and safety for its intended use, including securing any necessary regulatory approvals (e.g., FDA for Software as a Medical Device, where applicable) (FDA, n.d.b). Additionally, they must confirm the product's adherence to stringent data security and privacy standards, particularly HIPAA compliance (HHS.gov, n.d.; American Medical Association, n.d.b).

A critical step is obtaining informed consent, which necessitates comprehensive disclosure to the patient about the product, covering its purpose, benefits, risks, limitations, and costs. Providers must ensure that the patient's decision to use the digital product is entirely voluntary. Furthermore, meticulous integration into existing workflow and the patient's care plan is paramount. This includes planning how the digital product will seamlessly fit into current clinical processes, establishing clear protocols for monitoring and interpreting data, and defining how to respond to alerts (Calcium Health, n.d.). Adequate training for clinical staff on the product's use is also essential.

Providers are also accountable for ongoing oversight and evaluation. This involves regularly reviewing patient engagement with the digital product and assessing its impact on health outcomes. They must be prepared to offer technical assistance to patients and establish mechanisms for patient feedback. Continuous evaluation of ethical implications, such as potential algorithmic biases and equitable access, is also a vital responsibility (Frontiers, n.d.). Finally, thorough documentation is required, accurately recording the product recommendation, patient consent, and any relevant data or observations derived from the product's use in the patient's medical record.

PR-2. What are obstacles that prevent development, deployment, or effective utilization of the most useful and innovative applications for physician workflows, such as quality measurement reporting, clinical documentation, and billing tasks? How could these obstacles be mitigated?

Innovative healthcare applications, designed for critical workflows like quality reporting, documentation, and billing, consistently face substantial hurdles. Financial constraints are often primary obstacles, stemming from high development and implementation costs, uncertain returns on investment, and limited reimbursement models. This is particularly challenging for smaller or rural practices that frequently lack essential broadband infrastructure (Calcium Health, n.d.; Centers for Medicare & Medicaid Services, 2025).

Workflow integration and training burden also present significant issues. New tools can struggle to integrate seamlessly with existing Electronic Health Records (EHRs), leading to fragmented processes, increased administrative load, and resistance from staff (Calcium Health, n.d.). A persistent lack of interoperability and data silos continue to hinder smooth data exchange between disparate systems, resulting in fragmented patient data and inefficient care coordination (AudioEye, 2024). Furthermore, legitimate privacy and security concerns regarding sensitive health information, coupled with the complexities of achieving and maintaining HIPAA compliance, add considerable development and deployment challenges (U.S. Department of Health & Human Services, n.d.). Lastly, physician burnout, often exacerbated by burdensome EHRs, and a digital divide among patients with varying technological literacy and access, collectively limit the effective utilization of these promising innovations (Oxford Academic, 2025).

To overcome these barriers, a multifaceted approach is essential. Strategies must include clear financial incentives and clarified reimbursement models from CMS (Centers for Medicare & Medicaid Services, 2025), along with aggressive promotion of interoperability standards like FHIR. Designing tools with a workflow-centric approach that prioritizes usability and integrates seamlessly into existing clinical processes is crucial (MedCity News, 2025). Additionally, providing targeted training and ongoing support, implementing robust security measures, and offering clear regulatory guidance are vital. Addressing physician burnout through AI-driven

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automation of administrative tasks (Emitrr, n.d.) and bridging the digital divide through improved patient digital literacy and access are also critical steps for maximizing the benefits of these innovations.

PR-3. How important is it for healthcare delivery and interoperability in urban and rural areas that all data in an EHR system be accessible for exchange, regardless of storage format (for example, scanned documents, faxed records, lab results, free text notes, structured data fields)? Please address all of the following:

The impact of limited healthcare access on patients residing in rural areas is profound and complex, frequently resulting in significant health disparities when compared to their urban counterparts. One primary challenge involves geographic barriers and transportation limitations; rural residents often face considerable distances to healthcare facilities, particularly specialists. This necessitates increased travel time, higher transportation costs, and time away from work, which collectively impede access to both routine and specialized medical attention. Compounding this issue is the often scarce or absent public transportation infrastructure in these regions (Ricketts, 2020).

Rural areas contend with a chronic shortage of healthcare professionals, including primary care physicians, various specialists, and mental health providers. This scarcity leads to elevated patient-to-provider ratios, extended wait times for appointments, and a noticeable lack of continuous care (National Rural Health Association, n.d.). Economic disparities further exacerbate the problem; rural populations typically exhibit lower per capita incomes and elevated poverty rates. These economic conditions contribute to challenges with health insurance coverage and the overall affordability of care, often compelling individuals to forego or postpone necessary medical treatment due to financial constraints (Rural Health Information Hub, n.d.).

The issue is compounded by limited services and infrastructure within rural healthcare systems. Rural hospitals are generally smaller, frequently grapple with financial instability, and may offer a more restricted array of services, occasionally leading to closures. This directly affects access to essential services such as diagnostic testing, obstetrics, and emergency care (American Hospital Association, 2021). Beyond systemic limitations, socioeconomic and cultural factors also play a role. Lower educational attainment, higher poverty rates, social stigma associated with certain conditions (e.g., mental health), and privacy concerns stemming from a lack of anonymity in smaller communities can all serve as barriers to seeking timely and appropriate care (Lutfiyya et al., 2012).

Cumulatively, these interconnected challenges contribute to worse health outcomes in rural populations. These areas often experience higher incidences of chronic diseases, including cardiovascular disease, cancer, and diabetes, along with elevated rates of unintentional injuries. Consequently, rural residents frequently exhibit higher age-adjusted death rates compared to urban centers. A critical aspect of this outcome is the tendency for patients in rural areas to delay seeking medical attention until their conditions have advanced, leading to more complex treatments and generally poorer prognoses (Centers for Disease Control and Prevention, 2017).

2.1.2 Data Exchange

PR-6. Is TEFCA currently helping to advance provider access to health information?

Providers that connect from rural areas of the country and those that need to connect to many payer electronic end points would benefit from the capabilities provided by the QHINs under the TECFA arrangement and Sequoia RCA by reducing the set up activities used to link a single provider end point to many payer end points.

PR-7. What strategies can CMS implement to support providers in making high-quality, timely, and comprehensive healthcare data available for interoperability in the digital product ecosystem? How can the burden of increasing data availability and sharing be mitigated for providers? Are there ways that workflows or metrics that providers are already motivated to optimize for that could be reused for, or combined with, efforts needed to support interoperability?

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Innovative digital health applications for physician workflows like quality reporting, documentation, and billing face significant hurdles. Primary obstacles include high financial costs, uncertain return on investment, and inadequate reimbursement models, particularly for smaller or rural practices that often lack necessary broadband infrastructure Workflow integration challenges, and the burden of staff training disrupt existing clinical routines, leading to resistance and increased administrative load (Healthtechprime.com, n.d.). Pervasive interoperability issues and data silos prevent seamless data exchange between disparate systems, fragmenting patient information and complicating care coordination (AudioEye, 2024). Furthermore, privacy and security concerns over sensitive health data, coupled with complex HIPAA compliance, add layers of risk (HHS.gov, n.d.). Finally, physician burnout, often exacerbated by burdensome digital tools, and a digital divide among patients with varying tech literacy, limit the effective adoption and utility of these innovations.

To mitigate these obstacles, CMS can implement strategies focused on strong regulatory enforcement of FHIR-based APIs and information blocking rules, alongside strengthening health IT certification requirements to ensure native interoperability. Financial incentives must be expanded, integrating interoperability metrics into value-based care programs and offering targeted grants to cover technology and training costs. Providing extensive technical assistance and fostering awareness and trust by showcasing successful implementations and clarifying security measures will also be vital. To reduce provider burden, CMS can promote automation of data exchange through AI/NLP for documentation and billing, advocate for workflow-centric design that minimizes manual entry, and support the creation of a "single source of truth" for patient data. Crucially, CMS can leverage existing provider motivations by linking interoperability efforts to desired outcomes. For example, promoting digital quality measures (dQMs) automatically enhances interoperability as providers optimize data for reporting, while streamlining burdensome tasks like prior authorization through FHIR APIs benefits providers by improving efficiency and reducing rejections. By strategically aligning interoperability with existing clinical and financial incentives, data sharing becomes an integrated part of efficient practice rather than an added burden.

2. *Digital Identity*

PR-9. How might CMS encourage providers to accept digital identity credentials (for example, CLEAR, ID.me, Login.gov) from patients and their partners instead of proprietary logins that need to be tracked for each provider relationship?

2.1.3 How might CMS encourage providers to accept digital identity credentials from patients and their partners instead of proprietary logins that need to be tracked for each provider relationship?

CMS can take several strategic steps to encourage healthcare providers to accept digital credentials from patients and their partners. First, CMS can establish national standards and accreditation for digital identity by endorsing or helping to define a consistent and secure framework for federated digital identity, such as those compliant with NIST SP 800-63-3. This would mirror the approach taken by the solution implemented by Peraton, using the Okta Commercial-off-the-shelf (COTS) product, for accessing Department of Defense (DoD) applications. The tool is a secure, user-friendly way to access DoD applications, allowing eligible individuals and beneficiaries affiliated with the DoD, a single sign-on (SSO) to log into partner applications. Peraton's solution can federate with multiple digital identity providers, promoting an interoperable identity framework that could be used across Medicare services for providers and beneficiaries.

Second, CMS could incentivize adoption through reimbursement or participation models, similar to how the Meaningful Use program drove EHR adoption. Providers who accept digital credentials from trusted sources—such as ID.me, Login.gov, or healthcare-specific equivalents—could be rewarded for their participation.

Third, CMS can help reduce administrative burden by demonstrating how federated identity systems streamline workflows. For example, Peraton’s SSO solution reduces credential fatigue by eliminating the need to manage multiple user databases, handle password resets, or respond to frequent help desk requests, potentially leading to operational cost savings.

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Lastly, CMS can support identity federation through policy guidance and technological support, encouraging the use of a federated Identity-as-a-Service (IDaaS) model. This approach allows healthcare providers to connect to a shared authentication infrastructure, eliminating the need to develop or maintain separate login systems.

2.1.4 What would providers need help with to accelerate the transition to a single set of trusted digital identity credentials for the patient to keep track of, instead of one for each provider?

Providers will likely need assistance in several key areas to successfully implement federated identity services. Technical integration support will be crucial, as providers will require help integrating with federated identity services through APIs, single sign-on (SSO) protocols like SAML and OpenID Connect, and attribute exchanges. CMS can facilitate this by offering implementation toolkits, comprehensive technical documentation, and integration grants. Additionally, education and training initiatives from CMS can inform providers about the advantages and mechanics of federated identity, including how Attribute-Based Access Control (ABAC) can provide enhanced, role-specific access to patient data. To address concerns about HIPAA and cybersecurity liability, legal and compliance guidance from CMS will be essential; this could involve publishing guidelines on how federated identity aligns with privacy laws and offering safe harbor policies for providers accepting trusted third-party credentials. Finally, change management and user experience design support will be vital, as transitioning from proprietary logins to federated identity will necessitate workflow redesign and introduce patient onboarding challenges. CMS can assist by funding user-centered design pilots and providing change management resources to ensure a smooth adoption process.

2.1.5 How might CMS balance patient privacy with convenience and access to digital health products and services that may lead to significant improvements in health?

CMS can implement a tiered, attribute-based approach to enable granular access to patient data based on specific attributes like condition, eligibility, or consent status, rather than an all-or-nothing approach. Furthermore, CMS can use consent-driven federated identity models, integrating explicit patient consent flows into health service access to empower patients with control over their shared information, much like how federated identity systems incorporate user approval during login. To bolster privacy, CMS can also promote Privacy-by-Design frameworks, supporting the development of privacy-preserving authentication technologies such as zero-knowledge proofs and minimal disclosure tokens that verify identity without fully exposing credentials. Lastly, CMS can align with interagency efforts by partnering with the ONC, VA, DoD, and HHS to establish reliable and consistent cross-agency identity services for patients.

Peraton's SSO solution with the DoD, serves as a strong case study for CMS, demonstrating how federated identity, a seamless user experience, attribute-based access, and interagency coordination can lead to scalable, secure, and inclusive authentication systems. By adopting these principles, CMS can spearhead the transition to trusted, reusable digital identity in healthcare, effectively balancing privacy, efficiency, and accessibility for millions of Americans.

PR-10. Regarding digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 CSPs):

- **a. What are the challenges and benefits for providers?**
- **b. How would requiring their use improve access to health information?**
- **c. What are the potential downsides?**
- **d. What impact would mandatory credentials have on a nationwide provider directory?**
- **e. How could digital identity implementation improve provider data flow?**
- **f. Would combining FHIR addresses and identity improve data flow?**

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Carefully architected, UX design is a critical success factor for a large and diverse user community like those in CMS' provider and beneficiary community. Using adaptive, phishing-resistant Multi-Factor Authentication (MFA) in support of Zero Trust Architecture, Peraton's solution is supporting populations for DoD SSO including: service members, military recruits, current military students, former military students, Presidential Appointees, dependents, Veterans, spouses, family members, wounded warriors, care givers, retirees, civilians, contractors, vendors, foreign nationals, NAF employees, widowed spouses, and divorcees.

D. Payers

PA-1. What policy or technical limitations do you see in TEFCA? What changes would you suggest to address those limitations? To what degree do you expect these limitations to hinder participation in TEFCA?

Policy aspects should be considered when healthcare entities exchange data, they often enter into legal agreements, sometimes referred to as Trading Partner Agreements (TPA) which encapsulate the legal boundaries for each organization with regards to operating appropriately, often describing the practices which are allowed or considered infractions of the agreement upon execution. These infractions can be unauthorized data disclosure(s) which require one of the parties, possibly a government agency, to investigate the data disclosure and escalate actions appropriately up to and inclusive of criminal prosecution(s).

While TEFCA references 45 CFR 304 and 45 CFR 504 to address the disclosure(s) aspect, ASTP/ONC and CMS can educate the healthcare industry with emphasis much like the HHS/OCR privacy and security education efforts for the healthcare industry. Enforcement efforts to avoid the data breaches that have occurred recently in the industry must carry the penalties to hold accountable bad actors misusing or disseminating data. Simply providing credit report monitoring is insufficient to compensate the impacted patients when their data is breached.

Technologically aspects for payers of varying IT capabilities should account for more than the "lowest common denominator" level but bring the community to the same technological understanding and improved capabilities. Facilitation of this may require a shift from voluntary to mandatory participation through incentives which call for industry participation short off requirement through regulation. Learning from the early HIPAA process, unless there are conclusive regulatory requirements, the healthcare community will not make the investment of advanced processes and technologies to benefit the patient in the healthcare ecosystem without incentives or compliance requirements in place. With appropriate incentive type initiatives, healthcare stakeholders may be amenable to investing in upgraded technologies.

PA-2. How can CMS encourage payers to accelerate the implementation and utilization of APIs for patients, providers, and other payers, similar to the Blue Button 2.0 and Data at the Point of Care APIs released by CMS?

Commercial payers have created patient centric smartphone applications which allow the patient to see claims information after payment has completed. Larger insurance carriers have the resources to build these proprietary products due to their IT staffing. However, the information contained in these products may not be transferable to a new payer should the patient change which organization covers their healthcare transactions. Prior Authorization is one area where the use of Open Standards would contribute to the exchanging of patient centric information among entities. The CMS-0057-F Interoperability rule ignited the healthcare industry for the entities covered under that regulation. ASTP/ONC and CMS should consider balancing the drain on compliance with current resources building solutions to meet the needs outlined in the regulation and then adding the additional needs to build Blue Button 2.0 type solutions.

PA-3. How can CMS encourage payers to accept digital identity credentials (for example, CLEAR, ID.me, Login.gov) from patients and their partners instead of proprietary logins?

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ASTP/ONC and CMS can use the lessons and path forward that Peraton has used for other agency Single Sign-On (SSO) solution. They may be able to survey the current users of various digital products from various sources such as the Social Security Administration (SSA), United States Department of Veterans Affairs (VA), and state organizations e.g., those listed in the ID.ME government section, to gather problems encountered while these government sector entities established their credentialing processes.

Additionally, inquiring of entities using these credentialing directly (e.g., state agencies) to gather information for user maintenance aspects, version changes by each end point, time to update plans, to test, install and move to production, each of these products.

Peraton, with the Okta SaaS COTS product, has been able to accomplish this with a wide variety of users, and that experience can be leveraged for CMS' implementation as well.

ASTP/ONC and CMS can then begin an educational effort for the payer industry demonstrating the benefits of moving towards standards based digital credentials and away from proprietary identification methodologies.

PA-4. What would be the value to payers of a nationwide provider directory that included FHIR end points and used digital identity credentials?

ASTP/ONC and CMS can consider the currently under development HL7 Da Vinci Provider Directory effort as a potential starting point solution for the national "end point" directory. However, with the frequency of commercial organizations merging and acquiring other organizations, as well as those organizations which may exit their current configuration of healthcare industry support, should be considered as this will cause a ripple effect for the connections operationalized. Additionally, potential retesting of newer established connections may be required to avoid data leaks or breaches.

To protect patients from out-of-network expenses, updates of the provider information should be made within a short timeframe when contracted providers and payer networks negotiations end up with separation of providers from specific payer networks. Reinstatement actions must be completed and updated within one business day from completed and approved contracted negotiations between the provider entity and the payer.

As additional functionalities are discovered to be needed, the government would be well suited to address these needs with appropriate standards development organization(s) as to obtain a wider industry input when solving the needs of the healthcare ecosystem.

PA-5. What are ways payers can help with simplifying clinical quality data responsibilities of providers?

a. **How interested are payers and providers in EHR technology advances that enable bulk extraction of clinical quality data from EHRs to payers to allow them to do the calculations instead of the provider-side technology?**

   Currently, medical review operations are separated by claim type and often the patient requiring the service(s) to be reviewed. By keeping the data exchange to the minimum necessary both sides of the exchange (provider EHR and payer medical review) benefit by reducing excessive data dumping or extractions.

b. **In what ways can the interoperability and quality reporting responsibilities of providers to both CMS and other payers be consolidated so investments can be dually purposed? Are there technologies payers might leverage that would support access to real time quality data for healthcare providers to inform clinical care in addition to simplifying reporting processes?**

   CMS can establish a project scope statement with the standards development organization(s) to obtain input from other entities in the healthcare ecosystem which requires provider information for reporting

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purposes. This input can then be leveraged to standardize the data element(s) required for the comparable reporting requirements to each oversight organization collecting this data. Advertising such a project will attract willing participants and may also leverage common efforts that would be beneficial to all parties.

PA-7. How can CMS encourage payers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would it advance or negatively impact data exchange?

Payers have a certain leverage which providers do not enjoy about information blocking because providers need to meet the request for additional information from the payer in order to complete the adjudication process. Providers have a financial interest in completing all requests from payers for information and generally do act accordingly. Therefore, payers may have little experience with Information Blocking activities.

E. Technology Vendors, Data Providers, and Networks

1. *Ecosystem*

TD-1. What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?

While CMS has made strides in beneficiary interaction and provider interaction, there are still more inroads to be made. A large part of CMS initiatives has traditionally been focused on costs, payments, and intermediary's vs intuitive services that improve the interaction between the ultimate end users of CMS products, the beneficiaries and care providers. The simple fact that CMS has extensively communicated in recent months a renewed focus on proactive healthcare, Medicare beneficiaries and caregivers as a priority will serve to stimulate interest.

Medicare websites can be confusing and do not present well on mobile devices as Medicare recipients continue to become more tech savvy and prefer to use these types of devices to interact with various services. Medicare Claims processing is reliable, however still operating on legacy architecture that process claims in batch mode under preprogrammed payment timelines, that cannot take advantage of real-time data dissemination that would be beneficial to beneficiaries and caregivers. With constraints driven by internal bureaucracy, lack of knowledge on new and emerging technologies, and a complex stakeholder eco-system, there have been limitations that have constrained innovations and broader goals for large scale modernization. Understanding the Medicare eco-system, data, inter-connections and ultimately the end users, is a critical success factor to overcoming constraints of the past. The American public has grown accustomed to the immediate availability of Data and CMS lags in this regard across many of its functions. CMS can prioritize the modernization of these systems for API-First and real-time outcomes. Peraton has provided numerous options to consolidate and modernize the claims processing ecosystem to infuse new AI/ML technologies for the development of a new centralized claims processing system. Leveraging a phased approach will mitigate the risk and the new tools will enable developers to rapidly modernize. We look forward to the continued discussions with CMS on a path to real-time claims processing and a unified data strategy for claims.

A move to support native mobile apps to interact with CMS systems would be beneficial to increasing interaction with CMS beneficiaries and providers and with the added benefit of creating incentives to external developers to create products to facilitate this interaction. Moving to real time modern processing systems and eliminating data silos will also increase participation and innovation.

With access to beneficiary claims data and the move towards more proactive healthcare, CMS has the opportunity to leverage technology to provide proactive feedback on beneficiary health. This could be as non-

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intrusive as simple reminders to schedule preventive care appointments to more detailed health recommendations based on resultant healthcare visits. These notifications could be delivered through mobile applications, while protecting patient privacy.

Each one of these key changes in direction will create renewed interest and innovation on the part of the vendor community to develop new products to support CMS and its stakeholders.

TD-2. Regarding CMS Data, to stimulate developer interest--

a. What additional data would be most valuable if made available through CMS APIs?

Real or near real time data around Social Determinants of Health (SDOH) would be valuable through API access. This data often lags, where this typical population serves as one of the greatest utilizers of CMS services.

An additional dataset that would be useful is Realtime or near real time data around pandemics or outbreaks. While collaboration with CDC would be necessary here, having this data centralized and correlated with available CMS data would be beneficial.

Finally, to move to a more proactive Healthcare posture, CMS should consider the implementation of Universal Patient Health Record as an opt in option for Medicare and Medicaid Beneficiaries. While appropriate protections and protocols need to be put into place, as suggested in other portions of this RFI, to support such as system, both CMS and developers can leverage this data in mobile apps or online portals, or other messaging mechanisms to provide patient reminders, notifications, to push beneficiaries to seek preventive care as necessary to improve health outcomes.

What data sources are most valuable alongside the data available through the Blue Button 2.0 API?

As noted above centralized and correlation around SDOH, pandemics and outbreaks with available CDC data would be useful to vendor communities to innovate around healthcare digital products.

Additionally, Peraton has developed the Beneficiary Information in the Cloud (BIC) Application Programming Interface (API) which is a full suite API and toolset with the ability to onboard partners rapidly and disseminate beneficiary, claims, plan, and other CMS data to any trusted partner and could be scaled to deliver data directly to a Beneficiary facing mobile application.

b. What obstacles prevent accessing these data sources today?

Obstacles to access include the lagging nature or data. Near real time data is needed to make these data sources more usable. Additionally, the ability to provide correlated data across agencies (CMS, CDC) or even across CMS programs would prevent the need for vendors to consume available and perform data integration within their own systems or to purchase data from some of the existing Healthcare vendors that perform this function.

c. What other APIs should CMS and ASTP/ONC consider including in program policies to unleash innovation and support patients and providers?

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CMS and ASTP/ONC can establish a robust and secure ClinicalData API to enable providers to deliver patient clinical data to both patient applications and to CMS for streamlining data collection for audits, medical reviews, and FWA monitoring and investigations.

2. *Digital Identity*

TD-3. Regarding digital identity implementation:

See Peraton's response is section PR-9 for solutions on this topic.

a. What are the challenges and benefits?

Digital identity implementation in healthcare offers transformative potential but faces notable challenges for all stakeholders. For patients, key hurdles include the digital divide, limiting access for digitally underserved populations, alongside pervasive privacy and data security concerns that erode trust. The complexity of fragmented identity processes can also lead to frustration and hinder engagement. Healthcare providers grapple with the high costs and technical complexities of integrating new digital identity solutions with their diverse legacy systems, leading to workflow disruptions and requiring extensive staff training. Navigating the intricate and evolving landscape of regulatory compliance (e.g., HIPAA) further adds to their burden. Payers confront challenges in data standardization and quality from various provider sources, alongside the technical complexities of achieving interoperability across vast and diverse networks. Robust fraud prevention mechanisms must be developed, and a clear return on investment must be demonstrated to justify significant financial outlays for these new systems.

Despite these challenges, the benefits of robust digital identity solutions are substantial. Patients gain enhanced access and convenience, securely managing their health information and engaging with care services more easily. This also leads to improved care coordination, increased patient safety through more accurate data, and greater control over their personal health information, fostering trust. For healthcare providers, digital identity streamlines operations by automating administrative tasks, reducing errors, and cutting costs. It ensures more accurate patient matching, bolstering data security and compliance with stringent regulations. Furthermore, it is foundational for expanding telehealth and remote care, improving clinical decision-making. Payers benefit from improved member identification and data quality, leading to better fraud detection and reduced abuse. This translates into more efficient claims processing, enhanced member engagement, and more effective population health management strategies, ultimately fostering a more secure, efficient, and patient-centric healthcare ecosystem.

b. How would requiring digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 CSPs) impact cybersecurity and data exchange?

Requiring digital identity credentials compliant with NIST 800-63-3 IAL2/AAL2 standards would profoundly benefit healthcare cybersecurity and data exchange.

For cybersecurity, these high-assurance digital identities would significantly reduce fraud and strengthen access controls. IAL2 ensures rigorous identity proofing, making impersonation difficult, while AAL2 mandates strong multi-factor authentication, thwarting account takeovers. This creates a much more resilient defense against cyber threats and unauthorized access to sensitive health information.

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For data exchange, these verified digital identities would enable secure, reliable, and efficient information sharing. They instill confidence that individuals exchanging data are legitimate, facilitating seamless access to fragmented health records across disparate systems. This would streamline patient access to their data, improve data accuracy for providers (reducing errors and duplicates), and enhance payers' ability to identify members and process claims, ultimately leading to a more connected, efficient, and secure healthcare ecosystem.

c. What impact would mandatory use of the OpenID Connect identity protocol have?

The widespread adoption of OpenID Connect (OIDC) as a foundational identity protocol holds the potential to transform the healthcare industry, delivering substantial positive impacts. By its very design, OIDC would fundamentally streamline authentication processes, significantly enhance cybersecurity postures, and markedly improve the efficiency and security of data exchange and interoperability across the complex healthcare ecosystem. This shift would provide a standardized, robust, and user-friendly framework for digital identity, paving the way for a more integrated and secure future in healthcare.

3. *Technical Standards and Certification*

TD-4. How can CMS better encourage use of open, standards-based, publicly available APIs over proprietary APIs?

The use of open standards and publicly available APIs that is currently offered by CMS is step 1 to encourage organizations to move away from proprietary APIs. The value of the data that can be provided by CMS can also impact vendor behavior by eliminating the reliance on other vendors that correlate this data on their own. Continuing to communicate the value of these standards; interoperability, enabling innovation, enhancing data sharing, cost efficiency, can assist in the movement away from proprietary standards. Finally, CMS can implement policy whereby access to data is conditional upon its use, particularly in how that data may be shared, using open standards and publicly accessible APIs.

TD-5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?

A nationwide provider director of FHIR endpoints could significantly improve real time access to for the listed stakeholders. Such APIs would enable various mobile app functions such as provider search, geo mapping integration and even appointment scheduling for participating providers. Vendors could even use such data to correlate with social media postings or crowd sourcing to rate providers services and experiences which would be more valuable to beneficiaries, payers and providers than existing quality data currently published by CMS.

Assuming the data is enhanced based on some of the discussions above, it is reasonable to implement a cost on how this data is consumed to pay for the cost of collecting, aggregating, corelating, and publishing data sources. Multiple data vendors out their today provide this service at a cost and there is no reason as owners of the data that CMS cannot charge a reasonable fee to recoup the costs of doing this upfront work to enhance the value of its data to potential consumers.

TD-6. What unique interoperability functions does TEFCA perform?

The Trusted Exchange Framework and Common Agreement (TEFCA) enables a trusted framework to standardize and share health data across disparate health systems and provider networks. The TEFCA framework outlined by Office of the National Coordinator for Health IT (ONC) does provide a path to share data, however it requires the establishment of Qualified Health Information Networks (QHINs) to broker the connections and transfer of data. This approach will likely require significant investment by EMRs to develop a

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translator for packaging patient health data to the TEFCA standard, establishment of QHINs across the nation, and CMS and other HHS agencies to develop policies for leveraging the data. The TEFCA standard would not permit patients to access their data unless direct citizen access to QHINs is implemented, which could also introduce security and privacy concerns.

a. What existing alternatives should be considered?

FHIR or other API standards should be layered into this framework to allow patient access to their health data.

b. Are there redundant standards, protocols or channels or both that should be consolidated?

To ease burden on implementation TEFCA can permit FHIR based communications requiring limited changes to EHR platforms vs shifting immediately to a new standard.

TD-7. To what degree has USCDI improved interoperability and exchange and what are its limitations?

USCDI was a good step towards standardization of health data for interoperability. However, limitations include limited scope and currently the following data is not yet fully included; genomic data, imaging files, claims, billing, and mental health records.

a. Does it contain the full extent of data elements you need?

To fully enable beneficiaries to have full picture of their health record the missing data would need to be included, especially imaging and claims data.

b. If not, is it because of limitations in the definition of the USCDI format or the way it is utilized?

Limitations of current scope of data included in USCDI.

c. If so, would adding more data elements to USCDI add value or create scoping challenges? How could such challenges be addressed?

The health data is the most valuable of elements to add to USCDI, however the most significant challenges are to imaging data which varies in format for each medical imaging device manufacturer. A lower resolution standard may need to be adopted for sharing of imaging data and translators would need to be developed by the device manufacturers to support.

d. Given improvements in language models, would you prefer a non-proprietary but less structured format that might improve data coverage even if it requires more processing by the receiver?

This could be easier to implement, however many of the data formats are proprietary to health IT vendors and receivers may not have the same software to view the data, e.g. imaging data. As noted above for some data a standard may need to be mandated to allow sharing across platforms.

TD-8. What are the most effective certification criteria and standards under the ONC Health IT Certification Program?

If the goal of CMS is to improve interoperability, then TEFCA can be considered for addition to the electronic exchange category.

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TD-9. Regarding certification of health IT:

a. What are the benefits of redefining certification to prioritize API-enabled capabilities over software functionality?

Peraton has modernized several federal systems leveraging an API-First approach and the primary benefit that allows access to data must sooner in the modernization process. This approach will similarly center the importance in Health IT around the most valuable commodity, which is the data. Building systems around the data and the stakeholders that need it, prioritizing patients to directly access their health data, and sharing data across provider networks will all be advantages of an API-First approach.

b. What would be the drawbacks?

There are cost, provider burden, and health IT software development impacts, however these are not necessarily drawbacks as the return on investment for access to health data would be more beneficial to improved patient outcomes from standardized sharing of provider data.

c. How could ASTP/ONC revise health IT certification criteria to require APIs to consistently support exchanging data from all aspects of the patient's chart (for example, faxed records, free text, discrete data)?

A standardized Health Record format could enable health IT developers to develop converters for their data to translate to the new format. This would include structured, unstructured, and imaging data. The format would then need to be portable across FHIR and TEFCA standards to enable data sharing. Notable challenges would be on the imaging data as all medical imaging devices leverage their own standards, and a lower resolution file format standard may need to be adopted to share outside the provider networks.

d. What policy changes could CMS make so providers are motivated to respond to API- based data requests with best possible coverage and quality of data?

Peraton was the mission enablement contractor on the HITECH National Level Repository (NLR) program, CMS' EHR Incentive Program, which provided incentive payments for early adopters of EHR. A phased roll out of API data sharing requirements could be leveraged with a similar incentive approach, or conversely a penalty-based system on the provider claims should they not implement by a specified time. This, however, would be more dependent on the EHR vendors' ability to implement APIs than the responsibility of the providers themselves.

e. How could EHRs capable of bulk data transfer be used to reduce the burden on providers for reporting quality performance data to CMS? What capabilities are needed to show benefit? What concerns are there with this approach?

Peraton was the mission enablement contractor on the Physicians Value (PV) program which was one of CMS' first solutions for collecting quality data from providers. Enablement of bulk transfer of quality data (daily/weekly/monthly) could be easier for some provider EHRs. If the quality performance data is meant to adjust claims for some models, then there would need to be some standardization on timing of data collection.

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2.1.6 Data Exchange

TD-13. What new opportunities and advancements could emerge with APIs providing access to the entirety of a patient's electronic health information (EHI)?

As noted above, the largest benefit to this approach would be to allow CMS to be more engaged in proactive beneficiary care. Access to this data would provide even more analytical capabilities to shape programs and CMMI models. It would also allow the development of new applications that are patient centric, such as notifications or reminders for preventive care, or patient health dashboards that could integrate with Smart Health Devices.

a. What are the primary obstacles to this?

The largest obstacle to this approach would be patient privacy concerns. CMS could leverage an opt in model here or provide cost incentives to consumers to encourage participation.

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